

ActInf Livestream #006.2: "A tale of two densities" (2020) REUPLOAD

Livestream | 1:21:31

hello and welcome everyone
to the active inference live stream take
two
with an awesome set of flexible and
awesome participants
this is active inference live stream 6.2
on october 20th 2020
welcome to team com everyone we are an
experiment in online team communication
learning and practice related to active
inference
you can find us on twitter at
inferenceactive you can
email us find us on keybase or at our
youtube channel
this is a recorded and an archived live
stream so please provide us with
feedback
so that we can improve our work all
backgrounds and perspectives are
welcome here and also as far as video
etiquette for live stream goes
please mute if you're going to have
noise in your background
raise your hand so that we can get to
hear from everyone and use respectful
speech
today just as a quick point of news our
friend and colleague jared tumul has
released a very
interesting blog post called spinning up
in active inference and the free energy
principle
a syllabus for the curious and it is
a great place for beginners as well as

for those who want to go deeper into one
of the
various domains that are touched upon by
active inference and the free energy
principle
so today in active stream 6.2
we are going to have a warm-up we're
going to have take two
on our intros as i kick out this other
alex
we are going to then come to a
discussion of the paper
last time in 6.1 we talked a lot about
the goal of the paper
and we went through the roadmap and the
abstract in quite a lot of detail
this time we're going to be highlighting
some excellent follow-up comments from
our participants
we're going to be talking about pattern
versus structure metaphors
and more and then we'll go through the
figures
and make sure to talk about some of the
similarities and the differences
between bayesian networks and forney
factor graphs
great on to the intros and check-ins
so for this section please just
introduce yourself
and your location say hello provide a
little
short intro if you'd like and pass it to
somebody who hasn't spoken
yet so i'm daniel i'm in california
and i will pass it to our first time
discussant kate
i'm sorry i'm kate um i'm based out of

the university of edinburgh
i'm philosophy of cognitive fossil
cognitive science student and i'm
interested
in lifeline continuity and trying to
find a satisfactory definition
of what it is to be a living system and
i will pass
to mel
i think mel is not joined in yet let's
go to shannon
i think mel's here i don't know yeah
absolutely mel started uh
mel started her intro actually so mel is
absolutely here
we can hear you i don't know if daniel
can hear you yeah weirdly i don't hear
you mel could you just reload
sorry and shannon go ahead okay
no i don't think he sees you
this is a new and fun kind of just
reload the page mill
like just just click refresh yeah
exactly
perfect i see that mel take it away
thank you
there you go we're good yes yes
beautiful all right cool
i'm now andrew i'm in ohio i'm a
doctoral student in philosophy of
biology and company science
i'm interested in the same question as
kate and i will pass
to alex kiefer
uh hi i'm alex kiefer um i'm uh
in the philosophy department at monash
university currently in new york city
um i guess i'll probably be wind up

defending structural representationalism
about generative models today i don't
know
i'll pass it to maxwell i guess
i'll probably be agreeing with you in
the end uh i think alex
i'm maxwell ramstead uh i'm a
postdoctoral fellow at mcgill university
in montreal
i'm speaking to you from montreal right
now i work on multi-scale active
inference and i'll pass it
to i guess alex you haven't spoken yet
you're right bye we are
three alex um
well hello everybody um i'm
ali or alejandra um
i'm a professor in the department of
psychology specifically in the
department of um
cognitive sciences and well i'm
interested
in also the same question
that you were talking about what exactly
is to
be alive and
maybe going further what is cognition
and perception so yeah
um that's me oh well i'm i'm in mexico
um so i'm going to pass it to
um sasha
hi i'm sasha i'm based out of davis
california
and i'm a neuroscience graduate student
um
studying kind of the molecular and
developmental aspects of the brain
and i will pass it to shannon

hi i'm shannon i'm based out of uc
merced
in california but i'm in south dakota
right now i'm also interested in
how active infrareds can help us learn
about the brain
and music cognition um as well as how we
can learn about
how humans coordinate in music or other
sort of group interactions
and i will pass it to the other alex
hello uh i'm alex watkin
i'm a researcher at system management
school
in moscow russia i'm trying to find
possible ways of integration
active inference with system engineering
and system
management frameworks
so who's else cool i think that is
all of us and thanks everyone for
bearing with these
funny technical glitches i'm also
recording this locally so
worst case we'll just be able to have
this uploaded let's get on to
the warm-up questions so for the warm-up
questions just
anyone who wants to speak would be
awesome to hear from you
just to sort of get started what is
something that you've been
working on or learning about recently
whether it's directly linked or just
something that
influences your mind state coming into
today's discussion
and you can popcorn in and also i'll be

watching the stack if people
raise their hand
i'll start while people are raising
their hand we
just had complexity weekend this past
weekend which was a really
exciting and interesting time about 100
people
from 37 countries and forming cool teams
on

different topics really learned a lot
mel

hello yep that worked great beautiful
um yeah so recently so i've been i've
been sort of puzzling

for a couple years now god it's been
that one um

over how to conceptualize on the free
energy principle

um and so that's pulled me into
work in philosophy of science on models
and modeling

so so all different kinds of ways of
conceiving of

formal models and what they do and then
kind of parallels in uh in

biology in behavioral ecology
and ethology and evolutionary biology
with with models that sort of

[Music]

having some sort of empirical concept
cool anyone else want to share something
they're working on

or i can also just raise the second
warm-up question

just in case people want to answer that
one which is what is something that
you'd like to have resolved by the end

of today's discussion
whether it's a thought about the paper
specifically or
something general um sure
fellow gitzer i'm not sure that is
shannon go ahead
hey um i wonder
whether the uh like structural
representation
that we might be defending later today
um whether that structure could just be
a set of attractor
dynamics like is that enough to be a
structure
um so i think
alex might be raising his hand to have
something to say
yep alex oh yeah i mean i was gonna
speak anyway
i mean i would say yes to that question
probably
um it's not just my opinion but um i was
going to answer
both questions i guess so um apart from
trying to figure out the free energy
principles still
uh lots of conversations with people
i've been trying to um
i've been working up to trying to
implement a variational autoencoder to
figure out how that actually works
so um been doing a lot of sort of more
technical work um something i'd like to
have resolved is just
to figure out exactly where um
in activism or the version of an
activism that's defended in this paper
is inconsistent with representationalism

if it is
um so like it's i don't need there to be
residual disagreement but i just want to
figure it out
ooh sounds good any last thoughts on
these warm-ups before we
jump into it we have a lot of very
interesting slides kate go for it
yeah i guess the biggest thing that i
want to resolve by the end of the
session is i'm still not
really clear on the status of the
generative model because at some points
it sounds as if
that's something that's almost causal
that it's controlling the animals
behavior or
the organism or the system's behavior
and at other points it's
a model it sounds more like it's a model
that we use to describe the systems
dynamics rather than
something that is including beliefs or
being causal and that's something i'm
still not entirely clear on
so i think that's a big thing i won't
resolve
cool all really interesting topics
thanks for sharing and we're gonna
jump into it because a lot of these are
on the slides ahead
so the paper that we're gonna be coming
back to today is a tale of two densities
active inference is an active inference
by ramstead kirchoff and fristen
and just to rehearse the goal of this
paper they aimed to clarify
how best to interpret some of the

central constructs that underwrite the
free energy principle
and its corollary active inference in
theoretical neuroscience and biology
namely the role that the generative
models and the recognition densities
play in this theory
aiming to unify life and mind and so
last week we talked about what these two
densities are
and we talked about the tail that
integrates the densities
we also walk through the road map how
they build from a to z
by talking about statistical models and
representations
introducing active inference and then
combining that with
an active inference in this sort of
inactivism 2.0
framework before building towards
multi-disciplinary research heuristics
for cognitive science
so that was last week let's get into the
new and exciting parts
so this was one of the feedback comments
from a participant
last week and what they wrote was a key
learning goal for me in building my
understanding of active inference is to
develop intuitive examples for the
critical concepts that can act as the
basis for abstraction later on
it seems to me like there is probably
something like a logical sequence in
which this should be done but i don't
necessarily always know
which concepts and ideas from active

inference to prioritize

as an example coming from a cognitive

linguistics background i tend to think

of beliefs as either causal reasoning

or the underlying sensory motor

simulations that connect to linguistic

concepts i've been trying to find new

intuitive examples that can help me to

understand beliefs as probability

distributions as clearly this is the key

for making progress in understanding

active inference

so my question for all of you is what is

a system

or a metaphor that helps you understand

active inference and why

and while people are raising their hand

on the bottom we have just a few of the

systems that we've been discussing

over the last weeks we have a brain the

technical and the artistic parts of the

brain we have

insect colonies we have dance and

embodiment and movement

and then on the right i guess it

reflects something like the economy or

changes in dynamical systems but what is

a system

that helps you understand through

specifics

something about active inference

any thoughts on this

i mean i hate to be that guy but like i

started

understanding active inference when i

went into the math

uh like it's it's a formal theory and i

i think we can actually be misled by

uh metaphor uh

yeah uh some of my some of my

reservations about the philosophical

work that's been done on active

inference is that

it it operates on the basis of you know

uh kind of unformal

intuitive conceptual understandings

rather than trying to

see what the math that says so i hate to

be that guy though

no that's perfect we'll go to alex

bjotkin and then to

kate go ahead alex

yeah for me most important changed

my views was taking seriously

approach from like for process

and understanding

[Music]

concepts about states and steady states

of possible processes and the later on

all this concept with a markov blanket

as a border

and underlying structure like

generative models in links with

cybernetics it's bringing me

some kind of top-level picture and now

i'm trying to

focus on different parts to have more

detailed understanding

cool kate and then mel

yeah i was also going to say cybernetics

i think i only really started to

understand what was going on once i

started seeing the links with like ashby

and account of survival as stability

um that was when things started to make

sense for me um

and there's also just like a nice
metaphor i quite liked and i think it's
really
it's a paper by ellie rudenberg i think
with the union kingston
rightfeld with the crooked scientist
metaphor the way the scientist is trying
to
obtain the result that it's looking for
so it has to have some sort of
latch on the world in order to bring
about that result but it's still trying
to obtain
the result that it wants in the first
place rather than passively
find out what will happen i thought it
was quite a nice metaphor
so do you think they are helpful cool
mel
so i remember i'm i'm remembering um
my first sort of aha moment with the fpp
and active entrance
um i was i was explaining this stuff
to my friend who works on on similar
things but in a very different framework
um in the mit media lab and we had like
papers
strewn everywhere and scribbling all of
them and
my realization in explaining this
framework to him was that
um that there is
fundamentally at the at the lowest level
at the
at the level of the simplest sort of
systems or at their
kind of lowest scale right um
no difference between belief

updating and action so
a change in the system is a change in
the system
right and that is both
a belief update and an action right
um and once you build in
further levels of hierarchical
organization and complexity you get this
strong differentiation between them but
at the lowest level of this the same
thing

[Music]

cool and kate did you know that's a
probability
okay had you raised your hand there and
also i'm just
going to yeah alex kiefer go ahead
oh um yeah i mean i was gonna
that's i like that point that at least
at the lowest level we'll leave updating
an action or kind of
same thing i guess um that's one reason
that's one reason that i'm skeptical
that we need to explicitly i think
implicitly
more traditional bayesian brain stories
kind of
are built on that claim as well but um i
was going to say what got me into what
got me into active inference also
sort of a technical point was just that
you really can't you can't optimize the
thing you can't minimize surprisal
just by ingesting adjusting and
at least not okay the surprise depends
in part on the sensory observations
right so you can't minimize that
just by changing your generative model

um you have to
change the observations so that's that's
the technical thing that got me into it
um cool and yeah
thanks everyone for being flexible and
fun on the tech maxwell i really
agree that the metaphor as always with
metaphors it's a question about
compression of semantics and meaning and
at the end of the day we're talking
about something specific
and so we can think about different
systems that help us understand
how active inference is deployed in the
real world but often metaphors will take
off
take us off the mark uh because it
removes us from the real
underlying details so let's return and
the details are really where the devil's
at
uh in this case uh i think
but yep i i agree we're hopefully in the
end with the figures we're gonna be able
to look at those details really
specifically and i think that the way
that figure two three and four
lead into each other helped me
understand a lot about the system
so just to take another look at these
two
cities the two cities that are being
linked the densities
on the left we have the bayesian
structural representationalist
and alex freely added any details here
but bayesian models move between the
data parameters the observations and

hyper parameters or higher order
parameters and
from the data to the hyper parameters we
call it a recognition
model and to go from hyper parameters to
a set of data or observations that is a
generative model
and the outcome is a statistical
convergence of a multi-level model that
represents structures of the world
for example through an expectation
maximization scheme
we can contrast this with the inactivist
school of thought
which has linked rather than data and
hyper data
but rather has focused on the world and
agents and how they're linked through
perception and action
so a lot more adjacencies to areas like
niche construction
and ecological psychology and we
remember hearing from maxwell last week
about how it was the desire to
mathematically rigorize
some of the inactivism that led him to
the bayesian approach
and let's now think about how these two
densities are linked with a little bit
more feedback from one of our
participants they wrote the domino
metaphor finally makes sense
the physical dominoes are the physical
state of the system
corresponding to the recognition model
and the falling of the dominoes
is like the process or the dynamics of
the system reflecting the generative

model so here we've kind of combined
these two
different ways of thinking and we're
looking at the relationship
of the world and the agent or the system
and the system surroundings
and directly went ahead and combined
perception with the recognition model
and action with the generative model and
then also just to combine that with what
kate was just talking about with
cybernetics

this is a quote from the paper that says
on this view active inference can be
read as a new take on the good regulator
theorem proposed by control and rosh
ashby in 1970.

active inference tells us about the
relation between a control system
the generative model with priors over
action policies
and a system being controlled the
organism and its adaptive behavior
the actual actions undertaken in and
part of the world

so what what exactly does this clarify
about active inference
or what do people think about this
metaphor um or at least way to think
about how

if not necessarily a metaphor how would
we apply active inference to this
system what are we getting at here with
these similarities and differences
between the two
models

sure mel well so the good regulator
theorem the idea

is that um any good
control structure for some larger system
has to be a model of that system it has
to contain
in it all of the the key variables
for that system and basically what this
says to me about
active inference is that um in order for
a system in an environment
to be able to adequately react
to whatever kinds of perturbations are
happening in the environment
it has to contain all of the key
existence i call them existential
variables
the the variables in the environment
that would lead to the system
continuing to exist or not continuing to
place
cool and so how exactly i'm just curious
um does the system come to embody all
the key variables of the outside world
like don't the models of the organism
reflect a simplified
version of the outside world otherwise
there kind of is a
abortion story about the map that
represents the territory exactly so how
does a good regulator arise in the
context of an environment like that
um shannon and then maxwell
well i don't think the key is to be an
exact
like imitative model of the world or an
exact map even of the world
but just a model that is good enough to
enable you to
act and adjust in the world

cool maxwell and then mel
yeah i mean so formally speaking there's
a difference between
external states and hidden states um
and i mean when everything is going well
uh the external states that are modeled
by the system coincide with
the hidden states that are actually out
there
um and this is more of this it's not
i'm not i'm not sure if it's guaranteed
but
it it is strongly implied by the fact
that we're minimizing
uh variational free energy uh
that you know if if we're if we're not
generating a lot of
free energy then uh the external beliefs
that we have
about the environment tend to reflect uh
you know the causal structure of that
environment
but i mean manuel baltieri and chris
buckley have done some interesting work
showing that actually
the there isn't a necessary kind of
entailment relation
um you can have uh
you can have the the system can be
acting
that don't actually exist and i mean
this makes sense also
just evolutionarily uh perceptual
systems don't track truth they enable
adaptive
uh behavioral loops so if you're a prey
item like a bunny
it's adaptive to generate a lot of false

positives

uh compared to you know allowing for

false negatives and in terms of like

predator detection

it's it's more advantageous to you know

make a few

less costly mistakes than make a big one

so um

i mean yellow brinenberg and eric rivel

make make that point in the uh paper

that kate was

mentioning just earlier but uh yeah the

it doesn't actually have to hook up to

anything

in the real world though usually it will

end up doing so

cool mel and sasha

yeah so the the key term in what i've

said is

is key actually um so the key variables

right that's that's just what keeps the

system alive

can imagine that we first get a system

on the scene

that needs just basically one

parameter in its environment to be

correct in order to continue to exist as

a system

and then we build up from there to

systems that are progressively

more and more complex and need

progressively more and more

complex sort of environmental scenarios

to survive and then they themselves in

order to survive in those

more complex regimes need to build in

complexity

right in in terms of their their the

models
of the world that they enact right and i
think that says something
interesting about um how we
conceptualize uh
cognition and the onset of cognitive
complexity
um as sort of um and this is
peter garcia smith's line on the subject
is
is it's a response to a complexifying
environmental circumstance
right cool sasha then alex keefer
um yeah uh one key phrase that really
helped me
better understand um active inference
um is uh good enough and
that's what really um kind of put it all
together that
um while we're going about and reducing
uncertainty
about our system it just has to be good
enough to make the next
action or in the evolutionary sense to
survive
and that metaphor has really helped me
uh go through all this um all these high
level concepts
so thank you for mentioning that alex
kieffer
then alex fiatkin thanks um
yeah i just wanted to just note that
there's there's you don't need
to contrast accuracy or you know
answerable to the truth-ness with um
enabling adaptive behavior uh as as like
a dichotomy
um i'm not sure if that's what was

intended or not but this one reason i've
had trouble getting into
understanding the uh sort of
anti-representationalist
viewpoint is that um you know structural
similarity is a matter of degree
accuracy is a matter of degree in this
sense so um
i mean yes your hidden state
representation doesn't have to map
completely
accurately onto reality doesn't have to
capture every detail but if it doesn't
do that to any degree
um you're screwed so you know um maybe
we need
systematically um sort of simpler or
biased representations in order to do
the job but i think there's still a
implied relationship to the truth
there cool alex
yeah thanks uh i want to change
the system's level of consideration of
possible application of active inference
and for me interesting
on a personal level in terms of
day phenomenology and especially
professional and or a working day
phenomenology
how it works what is a generative model
and like an example
and what i won't discuss if i get it
correctly or
how it's good possible develop to think
about it
but if for example some doctor
he know different disciplines
and with different ontologies

and when he met a patient he
started his actions by his generative
model which activates with
recognition models to serve
as a doctor depends on exact situation
and exact case
and if it's so from for example from
another
level from team level if person
uh behave professionally in some
discipline
it became like a
perception model for a generative model
of the team
and what does it mean in terms of
learning and education of team members
because
if we can how
some in some way to link recognition
models with ontologies
and starting to work with it more again
cybernetically it's possible could be
very interesting
at least for me cool agreed the the
ontologies that how we think about the
world
definitely influences our perception
and sasha do you have something else
otherwise okay shannon go ahead
um i was just going back to alex
keefe's point that
your model like there has to be some
truth relationship with the world and if
we're going
to the dominoes and the process needs to
be that the fall to make some cool
pattern
if there's an interruption like so now

this domino can't reach
the next domino like that's akin to a
disruption and
how truthy the model is of the world
and it disrupts the action that you're
able to take or it disrupts the
um you die instead of being able to
survive
because your model isn't accurate at
that stage for the processes
to continue happening to do the next
action or to
survive cool cool discussion
on this domino system let's talk about
counter factuals and also return to some
of the great questions alejandra raised
in previous
weeks this is from another participant's
feedback they wrote
what i'd like to clarify if possible is
the following am i right in
understanding the generative model as a
control system that is dynamically
instantiated between
an organism and an affordance in the
context and the recognition density as a
relation
between the sensory inputs and the
possibilities for action policies
encoded in the organism
if so how does this work when action is
instantiated by counterfactual
forms of cognition or simulation i.e
those that do not rely on affordances in
the current external context
and so on the bottom right there's a a
younger person
looking into the mirror or maybe vice

versa and seeing an older person and this is reflecting how not only is there a deep temporal aspect to our action in the current moment meaning that there's often affordances that aren't available to us at the second that we're actually making the decision but also that we think about future possible affordances that don't exist based upon the sensory data that we're getting or do they and then on the right side i have a few bacteria because this is like an example of a system that we might not think of as having a simulation or counterfactual based mechanism of action often bacterial behavior can be summarized quite concisely by just the short term gradients that it's ascending or descending so for anyone to raise their hand and chime in how how do counterfactuals play into this and how do we make sense of affordances that aren't in a current ecosystem but are still important for behavior maxwell well i just want to comment on the first part of that question um so i mean i think it's a bit simpler than that in terms of the relationship between the recognition and the generative models or densities um so your recognition model is just your post your posterior the all of your posteriors basically all of the

all of your posterior estimates over
states
and precisions it's like my best guess
as to you know what the value of all
these states are
um and i mean
you can think of that as uh the system's
current
physical state so under under the active
inference um
formalism the current physical state
of the organism encodes uh
basically the the specific parameters
of these uh beliefs over external states
the
the actual physical states of the system
encode these parameters which
so yeah you embody a best guess
the the generative model is the point of
reference for the free energy gradients
so you can think of one as like you know
uh
the inference model and the other is the
control
model in the sense that the inference
model or the recognition
density is sort of like my best guess
right now and the generative model
is essentially you know a model of
what my phenotypic
preferences are and uh it's the tandem
between the two that kind of gets you
behavior
as in um the generative model provides
the point
uh the point of reference uh for the
definition of the variational free
energy gradients i.e

how close am i um to
my you know desired sensory distribution
and the recognition model tells you
where you are effectively
so it's like where where i am where i
want to be and the free energy is sort
of the difference between the two
effectively
um so i mean the first part of that
question yes you're right in
understanding the generative model as a
control system
um the recognition density is just the
best guess it's the posterior
uh so i mean it yes it is harnessing
these relations that are being described
here
uh but that's precisely because it's a
posterior estimate
of uh you know states and precisions and
all that
cool and one more thought on that is
for an agent who knows about for example
how hammers and nails are related
through their internal models they might
look at hammers and nails and pieces of
wood
and think you know later on i could
build a house with this
and so it's actually a future affordance
but it's based upon sensory input at the
current moment
um cool i'm gonna just
move on to the next slide and just a
short little quote by
buckminster fuller because sometimes i
just have to
he wrote rope may not be much like water

but
not is like the wave and this is from
the synergetics framework
and it's getting at this notion that
there's a difference between structural
integrity
which is like the rope and the water
which are quite different they're both
static and stable
to pattern integrity and knots and waves
can travel
through a medium and so by virtue of
traveling through medium
kind of like that domino wave there's
something similar
about these traveling waves that goes
beyond the mere structural integrity of
their media and so i just a thought
question for later is really
is active inference a structural
integrity or a pattern integrity
i would say that it's a pattern
integrity and just bringing up this
notion that many far from equilibrium
systems
though their mechanisms are extremely
different for example the bacteria
and the person there's going to be
similarities because of the way that
they're needing to persist
from a cybernetic perspective enough
there
let's get to that question about
structural aspects of attractors
so here a participant provided feedback
and wrote
why can't the recognition and generative
models be considered as structural

representations so they're coming from a structural representationalist perspective

they wrote in which case the dynamics of the structure are taken into account as part of the representation

even if we're appealing to dynamical systems the states of any particular dynamical system over time make up the structural representation

take the lorenz attractor the lorenz attractor has a specific approximate structure around two attractor states but the states and the trajectory of any point are sensitive to initial conditions

so any two initial points will lead to different precise physical states and dynamics

but will still travel through a very similar trajectory which hovers around these attractor basins

the states and the dynamics of a set of points over time can be taken as a whole and be identified as embodying structural representations of the lorenz system

so this is something from the realm of chaos and complexity theory and the lorenz attractor is an attractor that describes a simplified system of equations describing the flow of fluid and it's a fluid of uniform depth and there are also parameters for the imposed temperature difference the gravity parameters the buoyancy the

thermal diffusivity and the kinematic
viscosity

and the simple physical analogy here is
like heating up a pot of water there's
heat coming from the bottom
and there's cooling coming from the top
and there's this churning happening and
it goes around these two different
attractor states

so in this case maybe alex
uh or anyone else who raises their hand
what is structure

and what is representation and if we
agree on what this system is then what
is this representation debate that's
happening

mel

mel unmuted mel

or maxwell go ahead all right okay let's
go to alex

yeah mel if you want to speak english
it's raised then we'll go to alex keefe
and they'll raise their hand

okay well i was gonna say hey let me
talk um i don't have much to say about
the lorenz attractor um it looks like
a i mean what's i'm not sure what's
supposed to be represented in this case
like it i have an intuition in the case
of dealing with like

a biological system because you know
we're

intuitively representing some kind of
environment i'm just not sure
i'm not sure how to begin with this
example

um right we've got the state space
and this nice graphical representation

of the states of this of the state space
but
unless we interpret this entire system
as representing something i don't know
what to say about
its sort of content
okay cool mel
i don't know what a recommendation
is and so i
for that reason i guess i think that the
question of
what a representation is and
what in the world is a representation is
is maybe too ambitious only because i
haven't solved it
um but there's a related question that's
less ambitious which is
what is the difference between a
biological
function and a representation
um and that i think is more manageable
in some respects
in phil bio mind cogsi
we um speak of the disjunction problem
so we speak of and i know maxwell like
hates this framing and thinks it's
updated and we don't need it anymore
the idea is that in order for
representation to be a representation it
needs to have the capacity to represent
and we have a similar idea in biological
function which is
that in order for a function to be a
function it needs to have
the capacity to malfunction right so you
need malfunction dab function you need
misrepresentation that representation
um but how do we differentiate

a function from representation

good question maxwell then alex

cool um well so regarding the question

what is the representation

we targeted a very specific account of representation

i mean in the philosophy of cognitive science a representation is basically an internal thing within an organism that stands for

some feature of the environment or the organism's own body

in a way that the organism can leverage

to you know do something interesting

we focus more specifically on the claim that

generative models are structural representations

so uh i mean this has

been an account that's been uh you know opium o'brien first worked on this in the mid 2000s

uh and then gladsyevsky and milkowski in the context of predictive coding

and then alex kieffer and jakob howie

uh you know built on that i think in a very interesting way

and you know from that point of view structural representation

is an internal structure that's internal to an organism that gets its representational content

uh because it stands in a relation of structural similarity to some target domain

in the sense that uh second order structural features of that target domain like the statistical

properties of
the domain are recapitulated or like
mirrored in
uh the the actual properties of the the
representations themselves
that's the first uh there's a structural
similarity first property
second property it's not just that these
uh
structures are in the organism they have
to be exploitable in some sense
so the organism needs to be able to use
uh the
content uh encoded in uh
the representation to guide intelligent
adaptive behavior
um and then there are these two other uh
points that i think are pretty minor it
has to be detachable
meaning that the system can use it
offline uh
like as in uh you know not not like
actually engaging with the environment
and it has to be able to
afford representational error detection
uh in a matter similar to maps
um so those are the main features of
structural representations according to
the
the philosophical account on offer and
uh in this paper we propose that
generative models are not structural
representations
uh precisely because they don't really
need these
the these properties don't seem to be
true of
uh the generative models um

i mean yeah it i i think alex and i are gonna still disagree about this even after talking about it for like dozens of hours at this point but um yeah uh the main the main reason why uh in this paper we argue that they're not is that the generative model isn't encoded in anything uh and this is like one of the main differences between more traditional bayesian brain architectures and active inference as dan was pointing out really early on uh in traditional architectures well the recognition and generative models are just inverse the ones of the others the the recognition model is a mapping from the data to states and uh the generative uh model is a mapping from states to data so it's really just the same kind of set of connections and everything but it's just like which direction are you considering it like from the top down flow or from the from the uh the top down floor the bottom up flow of information um in the active inference scheme it doesn't work like that uh the generative model is nowhere in the dynamics sorry is nowhere in the physical organism it's only in the dynamics so uh it's very much like these dominoes falling over uh the wave of falling dominoes

isn't present in any of the kind of
single dominoes falling over uh the wave
is
is is part of the dynamic you know
phenomenon
uh and it's it's the same with the the
variational free energy and the
generative model the
generative model just is not there any
at any
at any time slice uh it only exists
in so far as we consider a lapse of time
as the point of reference for the
free energy gradients um so from that
point of view it can't possibly be
a structural representation because it's
not encoded by
anything cool well that's that's the
point a lot there
let's go to alejandra melvin
okay we'll go to alex alejandra
i figured i figured you'd have something
there yeah well
i mean that was a really a nice
explanation of what structural
representationalism is so uh
i don't have much to argue about there
um
and in general i don't i don't want to
argue more than is necessary
you know that about me but um i think um
the reason i want to insist on some of
this stuff is just to for my own basic
sanity to just feel like i have an
understanding of what's going on with
this stuff right
um so the reasons despite i understand
i think i understand exactly why from an

active inference perspective

um you wouldn't want to say that the

generative model is a

is encoded let me just say at first i

think

the question of whether or how it's

encoded is maybe a slightly distinct

question from whether it's a

representation and the reason i think

that it has to be a representation

well there are many reasons here one

argument the recognition model is an

approximation to the generative model

right

and so that's one example of like a

rational or probabilistic relationship

between these models and i don't think

that

that makes any sense if the generative

model is not a representation also

um so that's how that works on directive

inference

i i agree that in in a more traditional

bayesian brain sense the recognition

model is trying to

approximate uh the generative model i

mean that's

that's essentially how it's working um

it's an approximate posterior

approximation yeah

it's an approximate posterior but it

isn't approximating

the generative model well it's

approximating the posterior under the

generative model right

so so right and both

both the generative and the recognition

models are concerned their

distributions in part over uh
i have to be careful here not the actual
external states necessarily
but over hidden hidden states i don't
know
how on earth you get those into the
picture if it's not a representation
that you're talking about because the
states are environmental states they're
not necess
right we have to be as you point out in
the paper we have to be careful to
distinguish the generative model from
the generative process
we can't just identify what actually
happens with the generative model
we i think we need to have this is one
of the fundamental questions that i
think an activism often is faced with is
how do we deal with yeah i mean mel
brought up misrepresentation
um so the fact that the fact the point
is
the the way that you see the world the
organism or creature system sees the
world is not necessarily the way it is
but we still have that distribution over
external states as
essentially part of the generative model
let me just say one more thing here
about
encoding so the reason i think if you
look under the hood a little bit so
the generative model in under active
inference as i understand it
can roughly be identified with the
non-equilibrium steady state density
right something like that um

so i i mean i would just say right the
dynamics
the dynamics sort of instantiate the
generative model i think that's a really
cool point and i actually think that
structural representationalists didn't
do quite enough to
emphasize the fact that the generative
model is used as a control system
but i think if you look under the hood
there will be some features of the
system like stable
structural features in virtue of which
it has that nest density
so in fact you will find some stable
thing that you could think of as
encoding
the generative model maybe at least
that's
why i still have you know i'm still
attached to my views on this
cool let's do alejandra and then mel
yeah um i think i i um
i agree with alex or where are
i'm kind of confused how
the generative model can be conceived um
just like this an active process
um really i was like
reading the paper all over again and
um i actually don't get it um for me it
was
um the recognition density is the
inversion of the generative model
so uh if it is not encoded
like anywhere it's i
i feel kind of uh lost there
um for me yeah these top-down
connections

this is the generative model talking
about the brain specifically
and the recognition density is
related with this bottom up connection
so you can recognize
um your best guess right so
if it is not encoded um
what can be said about these top-down
connections
um i don't know
maybe i can jump in here and just
provide some points of clarification
sure about the generation wall if that's
okay yep um
okay so uh is screen sharing activated
um you could share your screen
if you yeah you probably should be able
to
okay let me try this
okay so this is from a paper called uh
the graphical brain
and this is what the generative models
look like all right can everyone see my
screen
yep also we're going to be returning
look at this in the center
yeah so the generative model is defined
as a joint probability density over all
of the variables of interest in the
system
right so uh in this case uh the
variables of interest are the states
that we're trying to
infer which are here s the policies that
i'm trying to pursue
which is here π and the data that i'm
observing at any given time
step uh which is here denoted o um

just so everyone uh is clear time flows
from left to right so this is the first
state this is the second state this is
the third state
and essentially the model you only have
access to this data point these are
counterfactual data points
uh and the whole thing is updated
basically at every time
step um and so look it the the
generative model itself like i said is
just a joint probability density
it's like the probability of all these
variables connected with
and effectively and when we say that
the generative model is not
present in the system i mean that this
density here
this joint probability density is never
encoded anywhere in the system
uh because the generative model itself
is factorized so
basically you you take this joint
probability density and then using some
uh you know bayes rule and and other uh
you know
chain rule etc uh manipulations
you can write this joint density as
the product of a bunch of uh likelihoods
and priors
and it's these likelihoods and priors
that
are uh updated constantly as part of the
recognition
density so this this prior
uh you know about your your data given
your states and this other prior about
your states given the next date and the

policy and all this
these are updated uh dynamically
and their current value uh like all of
them together
can collectively comprise the generative
model
uh yeah that's right sorry the
recognition model
the uh generative model is how these
quantities are all connected the ones to
the others so like if you want to think
about it heuristically
uh you know this uh all of these
specific parameters like the s the π
the o
all that like there all of those values
together
as they're being updated uh comprise the
recognition
uh density and the generative density or
the generative model is really just
these relations between uh
the different quantities as they change
so
it's it's literally that the um
the inference process
itself is what holds all these
quantities together
and the generative model is just a
description of how those
quantities flow together uh
yeah and so again it's like the domino's
uh falling over
thing uh you know the the the wave of
dominoes only exists in that motion
and and similarly the generative model
only exists
in that kind of coordinated inference or

update

dynamics of the quantities that are part
of the recognition

density um yeah

and the reason it's called the

generative model at all is just because
it's borrowed

it's a terminology borrowed from machine
learning uh

in machine learning the the joint
probability density over all of your
variables is known as a generative model
so it's it's called that way

just because that's the that's the
technical

term uh it's it's also called that way

because if you're starting from a joint
density over all of your variables you
can actually generate

fictive data that you would expect under
this configuration

um of different uh parameters

oh thanks for that clarification maxwell
will have those middles

we'll have mel and then we'll carry on
with that all sounds pretty good to me
go for it mel

i think kate and mill have yeah yeah

okay kate and then mel oh i just wanted
to um

follow up to what maxell was saying
about structural representation

can you hear me yep go for it

can people hear me yes yes yes we can
hear it beautiful

awesome yeah just to follow up to what
maxwell was saying about

um structural representations uh

the the sort of detachability of the
representation right that's
that's what should get us the
distinction between
a biological function and at least a
structural representation right because
my feet and legs and
knees are a representation of
ground in some very minimal sense in the
same way that
fish fins are representations of
of fish fins and and bird wings or
representations of fluid dynamics right
um but this isn't
detachable from immediate environmental
circumstance for offline use
right um so that's what should get us
the difference between uh
function representation but i'm i'm
curious to
see what role that plays in
in the ftp in active interest um if any
do we want to do we want to is there is
there merit and retaining a distinction
between
function eruption patients under under
active infants in the sap or are they
just
sort of continuous with one another nice
very good question kate did you have
something there
okay okay i'm gonna just continue with
the slides because i'm not sure if i
think kate's up next
yeah oh sorry i wasn't first just wanted
to apply um i think i'm still
with the in the case of the the paper
was really useful for me in like

helping me understand what the generation model was in the fep almost all the way like when it's described as like a statistical description of the generative process the way you've explained it all of that makes sense to me um but then there's also parts in the paper where it says where you think it's like the generative model sort of um is what the organism expects and it guides what the organism does which makes it sound like the generative model is playing this kind of causal role not just for the generative model itself um and i find that kind of confusing if the given the generative model isn't something encoded by the organism um or by the system um that i'm still i get the idea of it as a statistical description of the dynamics that sounds right i just don't understand how in that how if that's all it is that it's something the organism uses it seems like it's something we use to describe the organism rather than something organism itself is in any way using um that's a good point i i've proposed in a recent paper uh that came out in entropy not too long ago that uh both are correct so the you basically the the correct way to interpret the fep

is as
instrumentalism next nested within
instrumentalism
meaning that we can have a model
theoretic
uh philosophy of science reading of the
fep
and you know for that matter any other
theory of cognition
as this is a model that we as scientists
are using to explain
the behavior of organisms
what i've been trying to argue is that
the fep itself at the theory level
is also saying the organism is
exploiting
the statistical structure of its body
and movement aka the generative model
to guide adaptive behavior so the model
really is these harnessed
relations between the variables that
have to
obtain for survival to persist
uh and you know so it's it's sort of
this idea that once the system kind of
gets moving then there's kind of a
an inertial dynamic thing
kind of keeping it moving so like you
know my
core body temperature is 36 and a half
degrees
and the fact that it is keeps a bunch of
other processes in play
that in turn keep me preserving my
temperature
uh you know by initiating adaptive
behavior like putting on a parka and it
gets cold and so on

so uh yeah i mean both
both kinds of generative models are used
separately in dynamical causal
modeling uh of the kind that the fristen
group
used to model the spread of covid we're
not assuming
that the process that we're modeling is
itself an active inference agent
um and it so you can do that
but you can also make the additional
assumption that it is
and then you're in the game of active
inference proper
cool thanks max i'll go ahead kate
can i come back to that go ahead go
ahead yes absolutely
i i the thing the kind of worry i get
there then it sort of seems
like you're losing the distinction
between the vehicle of the model and the
target of the model such that
i worry it sounds a bit like just saying
everything is a model of itself
essentially um i mean effectively that
is what we're saying
and that's where it gets a little hazy
for me like you know that's probably the
point of disagreement between alex and i
also i want to say the generative model
just is the organism
in the sense that like this this
specific you know statistical structure
that harnesses all of these different
you know existential variables as
mel was saying like this really just is
the organism
like this kind of statistical structure

that keeps kind of
reiterating itself and coming
into existence that just is the organism
and that's one of the reasons
why i don't want to say that this model
is a representation of anything
you could just call it you know the
phenotype
we call it a generative model because
formally speaking
the same construct is borrowed from a
field where in which it's called a
generative model but
you can think of it as the organism's
phenotype and precisely because it's a
bit weird to think of
the model as modeling itself i kind of
resist that interpretation
and say well you know if anything is a
model really in the more traditional
sense it's this recognition density
that's performing posterior state
inference
and i'm sure that alex will tell me why
i'm wrong
in a second it's it's a great
response let's just move on through a
few more of the slides because there's a
lot
in the figures that i think gets at this
um but mel has their hand though
sorry go ahead mel
oh i just want to i just wanted to
introduce it i think i think
maxwell is possibly mismodeling himself
i think he's
i think he's actually a tad bit more
realist

than he's letting on oh maxwell
a realist at heart on my bad days
exactly if you can't if you can't uh
take care of me at my most real
so in this next slide one of the
feedback just from a participant just to
give a
example and we won't spend too much time
here
but i like this idea that the beliefs
don't have to be verbal
uh that the physical therapy context
shows us how the range of movements that
the body does not
want to go to for example due to a
traumatic experience or injury
embodies a belief that doesn't involve
language and this was a paper with
uh limanowski and fristen from 2020
where a person's hand was opening and
closing and then they were getting a vr
visual that was delayed or was
inaccurate
and so there's so much here about how
our sensory feedback are proprioceptive
as well as our visual feedback
multimodal systems
are being used and that's the mess that
would take us you know so many hours
and more to clarify this is what the
representation questions are about
so it's like the person what is the
representation of the current state
of their hand being closed at that just
pure
phenotype level maybe somebody with a
ruler says well it's whether the hand is
closed or not

but it's not just that because if you're
getting visual input that's messing with
you
it changes your behavior so there's
something else happening and that's
really
a lot of what we're talking about um
i think going through the figures two
three four
are going to be really helpful before we
have closing
thoughts and thanks everyone for bearing
with this was sort of a
weirdly technically altered stream
briefly
the bayesian networks so bayesian
networks are
from bayesian statistical sciences
and in these diagrams the nodes are
variables they're like random variables
stochastic variables or parameters
the edges are statistical relationships
they're probabilistic relationships and
they can be undirected
or directed these models are quite
common in statistics
and also there's some really nice
aspects about them
in that you can bring prior knowledge to
the table with your priors and you can
also estimate it directly from the data
with a parametric empirical base
and to give an example about what this
looks like here on the top right
we can see how like an accident
increases the likelihood that you'll
hear sirens
an accident also increases the

likelihood of having a traffic jam
bad weather it can increase the
likelihood of accidents or traffic jams
so these bayesian networks are pretty
commonly used in a lot of statistical
areas
and we can contrast that with what are
called forney factor graphs
and working through this devries and
first and paper was definitely
quite an experience this is a complex
paper and there's a lot in it
but in a forney factor graph as they
write in the paper it's a type of
graphical model that shares qualities
with bayesian networks and markov random
fields
forney factor graphs afford a visually
insightful representation of generative
models
especially beneficial for complex models
underlying hierarchical active inference
in a forney factor graph each factor is
represented by a node
and each variable by an edge so it's
like a flip of the bayesian
an edge attaches to a node if the edge
variable is an argument of the node
and so here on the right is a table from
the paper of devry's and fristen
and it shows how we can think about
these functions like addition
subtraction multiplication as having
variables coming in
and out and just to really quickly
summarize the similarities and
differences
in the bayesian networks the nodes are

variables whereas in the forney graph
the nodes are the factors or the
functions
in the bayesian networks the edges are
the probabilistic relationships amongst
variables
whereas in the forney factor graphs the
edges are the variables themself
a cool trick about bayesian networks is
this model inversion
which is that the model can be gleaned
from the data
and also the data can be generated from
a model another cool trick of
fornigraphs is that they specify
a specific order for factorizable
computation
making extremely massive models
tractable and then a last cool fact
about a bayesian network is that's where
the markov blanket formalism comes from
which is it's the set of nodes that
insulates one part of the network from
each other
from a probabilistic perspective though
there's a there's a lot of nuance in how
it's used in the free energy principle
framework and the cool fact about the
forney graphs is that it's
linked to information theory and message
passing computation and other math
so in figure two we um see
this generative model in active
inference and this is actually the model
that maxwell had on his screen earlier
and i annotated all the pieces so just
as maxwell was saying time flows from
the left to the right

and in this is a again a bayesian
network representation
so the the boxes or the circles and
there's different types of them which
are described in the caption
the boxes and the circles are like
variables so the initial state is a
vector
the hidden states are variables the
state transition matrix is
how the hidden states map to each other
that's b
a is the map between the states and the
observations
and then figure three is the same thing
as a forney factor graph and so this is
showing that there's this equivalence
with representing something as a
bayesian net
and a forney factor graph and the forney
factor graph
allows for these extremely large scale
hierarchical models
to get factorized and then computed in a
tractable
way and so i should say like the
advantage of using a forney style factor
graph is that the message passing
is just you can just read it directly
off the forney style factor graph
and there's an algorithmic way of moving
from a bayes net to a forney style
factor graph
so typically intuitively the bayes net
is easier to write
uh because you can you know you can just
literally write down
the dependencies between all of your

factors

like this uh so you know like i don't

this state depends on this other state

and you know there are these transitions

and etc blah blah

and so you know from this you can move

to the forney style factor graph and

that tells you

just directly how that will be

implemented in uh

active inference i should say that the

reason why i we discussed these

at all is really just uh to show in the

next figure how the

in figure four how the generative model

and the generative process kind of link

up uh i mean to me this is where the

the interesting stuff happens where like

literally the the generative process and

this is a model of the generative

process right like how we think

it's structured it's in this little box

here um

and then the 40 style factor graph

represents the message

passing that is ongoing yeah outside the

box

uh to keep all the the quantities within

you know phenotypic bounds

thanks maxwell and yeah i know that was

a little bit of a rapid sprint through

these figures

but it's so cool that there's a formal

way

to relate the bayesian and the factor

graph

models and just as you specified it's

easy to specify sometimes the

bayesian network just like oh well the traffic jam depends on the weather and the weather depends on this other feature and then the factor graph allows us to have an order of computation and a really uh implementable way to compute it and also it has some similarities with message passing for example in neural network systems so it's it's an interesting question that i've been having is okay we talk about this is as a bayesian theory but if it can also be represented in another way is it is it really a uniquely bayesian theory or what do we gain by saying that it's a bayesian theory it's uh within the domain of bayesian inter-transformability uh or interoperability but that doesn't uniquely uh disambiguate it and that's why i think some of these philosophical questions that we're talking about here like about representation and about cause are so open because uh it's it's not uniquely specified by the free energy principle or by active inference like how exactly we should answer these philosophical questions because in the end as maxwell said it's the technical details this is what we're talking about when we're talking about the the free energy minimization and how that bears

upon the policy selection
this is what's happening and also one
more point before if anyone
raises their hand to provide a comment
is um
notice that the free energy is linking
to the policy selection
pi p for policy and that means that
the organism isn't just minimizing the
free energy of the observations given
a sort of neutral representation
actually the policy representation
specifically how it bears upon how
states transition into each other
that's what the actual computation is
being done
upon so the observations are being used
to run then
upstream through the hidden states
and their transitions into the policy
that ultimately dictates what's going to
be done
by being selected by the organism or the
system as the most
likely so this is really how deeply
action is embedded in these models
it's not that there's a latent
representation of the external states
and then you're doing a best guess on
what you should do it's actually
like action is at the top here the
policy is at the top
and so i think that's really a nice
feature of these models and it becomes
graphically clear
and then if somebody wants to have
philosophical framework a or
philosophical framework be

on top of this structure or if they want
to propose a different structure
for how observations and policies are
linked that's
something great it's something tractable
that we can talk about but
this also returns to the metaphor
discussion this is the real
thing as it is um cool i know we had to
sprint through there but it's just um
coming near the end of our time so any
thoughts on these images
or are people ready to sort of go to our
closing
thoughts i was just saying the equations
are more interesting but
you have to say the equations are
interesting and and
the 2017 to verizon fristen and this
model it just reminds uh me that if we
can write down that bayesian
relationship the probabilistic
relationship between variables
then there's going to be a process model
that is going to be how we can link it
and these captions are very informative
and there's really a lot to
say here but enough for today
let's close it out um we can hear any
last thoughts but
while people are preparing their last
thoughts and raising their hand if
they'd like i would just like to
thank everyone for participating we're
learning
and developing each time figuring out
the tech each time we
provide follow-up forms to the live

participants so just check the calendar
invitation
and it would be really helpful to have
feedback
we also request feedback suggestions and
questions
from the participants live in the
comments
or otherwise and uh just stay in
communication
we really appreciate everyone's
engagement here and it's been
such a great learning experience so if
anyone wants to raise their hand
alejandra first and then followed up by
anyone else who wants to
let's just have some closing thoughts
[Music]
well i was wondering if
if it's possible to like model updating
if
when you are doing offline cognition
um we like without like an explicit
policy you are not doing something
online
when this interaction bodily interaction
you are thinking about something and
then you have like this
uh like you know said like this eureka
feeling so this this is kind of a model
updating
um without action
um i don't know the short answer is yes
i just linked the paper to it we uh
we've created a ruminating agent
um so basically this agent combines the
affective
inference uh scheme that we

uh have developed over the last two years where you're effectively inducing a new hierarchical layer of state inference based on state and precision estimates at the lower level and these are affective states that then guide lower level precision estimation so we combine that with what's called sophisticated inference which is the new generation of active inference agents um you know vanilla formulation current vanilla um you could see that in the generative models that we just showed they include uh beliefs about counterfactual observations that we would make as we propagate states into the future sophisticated inference is about propagating your beliefs about the observations that you would make in counterfactual futures and so by combining these two things we have an agent that isn't acting uh but that ends up ruminating um and catastrophizing about uh its own cognition um so i just linked the paper in the chat um we we submitted it to the uh the international workshop on active inference uh so the it's a little poster now but we're we're slowly expanding that into a full manuscript but the the kind of summary six pager uh that just does what you were

suggesting ale
is uh this is it yeah
thank you maxwell um but this is kind of
confusing with the an active uh
view of cognition well this is why we're
not we're not like
uh classical and activists so yeah we
reject the equal partners principle
it's just not the case that uh the body
uh the brain and the cultural
like mealier are going to be involved in
the same way
for every task so like for rumination
you know you could you could i guess you
know appeal to a history of interaction
which is you know the reason why agents
are able to entertain beliefs about
other agents at all
and so on and so forth but yeah you're
right there isn't much
in action here and that's fine i think
uh
alex kiefer and then anyone else who
wants to raise their hand
uh yeah so i guess in lieu of a proper
closing
statement i just wanted to point out
that maxwell has some really cool new
work
on how the neuronal populations sort of
can be seen as an acting
uh inference on their own level and that
this is one way you could see
uh structural representations being sort
of realized so
like that one thing we didn't have time
to get into um but i just wanted to say
in general

like i think um yeah there is
i think regardless of the precise
philosophical take that you have on all
this stuff like
it's cool to disagree on that level but
this is this literature has produced
some really amazing
work and i think it's it's pushing
forward in a really interesting way and
um
one one advantage of active difference
certainly is that you can actually write
down the generative model and
and sort of transparently see how it
works and you don't get that out of just
unsupervised learning
who any other thoughts
awesome well thanks so much for the
discussion we're gonna
um re-watch understand technologically
what did
didn't work and for next week provide
all of you
plus anyone else who wants to
participate with more of a specific spec
sheet for how
you can prepare and make sure that it
will be a really like seamless
experience
and as far as the timing goes as well
next week we will move on to
active seven which is going to be about
variational ecology
which is a really awesome paper close to
my heart as a biologist as well and we
would really appreciate anyone who wants
to participate
in these conversations next tuesday at 7

a.m pacific or the following
tuesday both of these discussions will
be really great and it's quite a complex
paper so i'm sure we'll have a lot to go
into again so once again
thanks so much everyone for your
participation
really brought up some awesome ideas and
i hope we conveyed to those within
and without of the field that there's a
lot of discussion
ongoing by no means are these questions
resolved and it's an opportunity for
people to bring their unique
perspectives to the table
so thank you all again for your
participation
really appreciated and we'll see you
later